

Weekly 9

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Weekly Progress (5 points)

1. Evaluated outcomes from the previous week's QCA and the output statistics it gave alongside the conditions it found necessary to decide to pursue a bit more calibration.
2. Researched further on calibration for QCA and determined it would be okay and statistically sound to model the calibration off of theoretical understanding first and achieving a decent logistic distribution second.
3. Updated the QCA calibration to better fit the above considerations in order to have better outcomes from the analysis – this involved just a little while of testing all the variables with numerous slight changes to the calibrated values for exclusion, inclusion, and crossover point until it seemed to give enough cases for multiple levels.
4. Did further reading on the QCA package in R – specifically spent a decent amount of time parsing types. This was not *perfect*, and was motivated by some errors I'd encountered every now and again that were unclear, but it was helpful and will be useful for poster writing.
5. Updated methodology to allow for more information to be drawn from it – this involved manually making the truth tables and minimizing it (which `superSubset` did not do); for the noninternationalized case, this did make things substantially simpler.

Challenges (3 points)

1. I spent a while trying to see what I could do to make the calibration "better", which, even now I'm still kind of scared that the data is messed up to some degree and won't yield great results (though that's probably just me being anxious since I've been pretty careful overall). I guess one should probably get used to that being part of research before adequate consultation with a number of experts – it's been present when I've done it before. What I've done seems to be a good balance of political science theory and methodological interests despite the debate I've seen on it, so I can sleep well I suppose.
2. It would be nice to have a better understanding of the underlying processes for the R QCA package just to better understand what's going on at a code level, but between the lengthy GitHub page for it and the lack of external discussion, it just isn't worthwhile enough to spend a few hours on just that (at the present juncture).
3. One thing that I am noting – possibly from having too many variables or too few cases (changing out some data for a proxy measure that's more available can fix the latter) – is that there might not be enough combinations of factors that are at the desired significance level. Maybe that reflects reality, maybe it doesn't, but it just means I'll have a bit more to tinker with to see if I can do anything a little better (though this will impact my interpretation).

Code Commits (2 points)

1. Updated calibration to better fit desired distribution:
<https://github.com/GW-datasci/26-spring-LUTZ-H/commit/d703c15cdc64a65a1608a3488f8d5c8799a988ba>
2. Updated methodology for easier interpretation:
<https://github.com/GW-datasci/26-spring-LUTZ-H/commit/66db41a22124db8dcd569441172073383fed972f>